



United International University
School of Science and Engineering
Department of Computer Science and Engineering
Final Examination || Trimester: Fall – 2025
Course Title: Calculus and Linear Algebra
Course code: Math – 2183 Marks – 40 || Duration – 2 hours

[Note that the marks are given in brackets at the end of each question or part question. You have to answer all the questions in order.]

Q1. (a) Consider the following coefficient matrix of a homogeneous system of equations: [3]

$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{pmatrix}$. Find the **rank** of the matrix and show that the system of equations has infinitely many solutions.

(b) Consider the following system of linear equations:

[5]

$$\begin{cases} x + y + z = 3 \\ 2x + ky + 2z = 6 \\ x + y + kz = 3 \end{cases}$$

Write the system in **matrix form** $AX = B$. Find the values of k for which the system has:

- (i) A unique solution,
- (ii) Infinitely many solutions,
- (iii) No solution?

(c) The matrix $A = \begin{bmatrix} x & 2 \\ x+1 & 3 \end{bmatrix}$ has no inverse. State the value of x . [2]

Q2. (a) The roots of the characteristic equation of a linear ordinary differential equation are $-2, 1$ and $2 \pm i\sqrt{3}$. Write down the general solution of the differential equation. [2]

(b) According to **Newton's second law** and **Hooke's law**, the motion of a mass is governed by the differential equation: [3]

$$2 \frac{d^2x}{dt^2} + 8x = 0$$

Solve the differential equation to find the **general solution** for $x(t)$.

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- (c) A damped mechanical system is subjected to an external force consisting of an exponentially decaying term e^{3x} and a linearly increasing term $3x$. If the damping and restoring coefficients are both equal to 4, which form the differential equation. [5]

$$y'' + 4y' + 4y = e^{3x} + 3x$$

Find the displacement $y(x)$ of the system.

Q3. a) Consider the following matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{2} & -\frac{1}{6} \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 2 & 0 \\ 2 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix}$$

- i) Verify whether C is the inverse of A or not. [3]
- ii) Find $\text{tr}(B + C)$. [1]
- iii) Evaluate $\det(D)$ using cofactor expansion. [2]
- iv) Find D^{-1} . [3]

Q4

- a) Find eigenvalues and eigenvectors of the Matrix $A = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}$. Hence; [3]
 - i) Show that, $\det(A)$ = product of eigenvalues. [1]
 - ii) Confirm that, $\text{tr}(A)$ = sum of eigenvalues. [1]
- b) Triangle ABC has vertices $A(1, 1), B(3, 1), C(2, 3)$. Find the image of the vertices when triangle ABC is
 - (i) Rotated 90° clockwise about the origin. [2]
 - (ii) Enlarged by scale factor -3 and center origin. [2]
 - (iii) Reflected along $y = x$ line. [2]